

Project Design Phase-II
Data Flow Diagram & User Stories

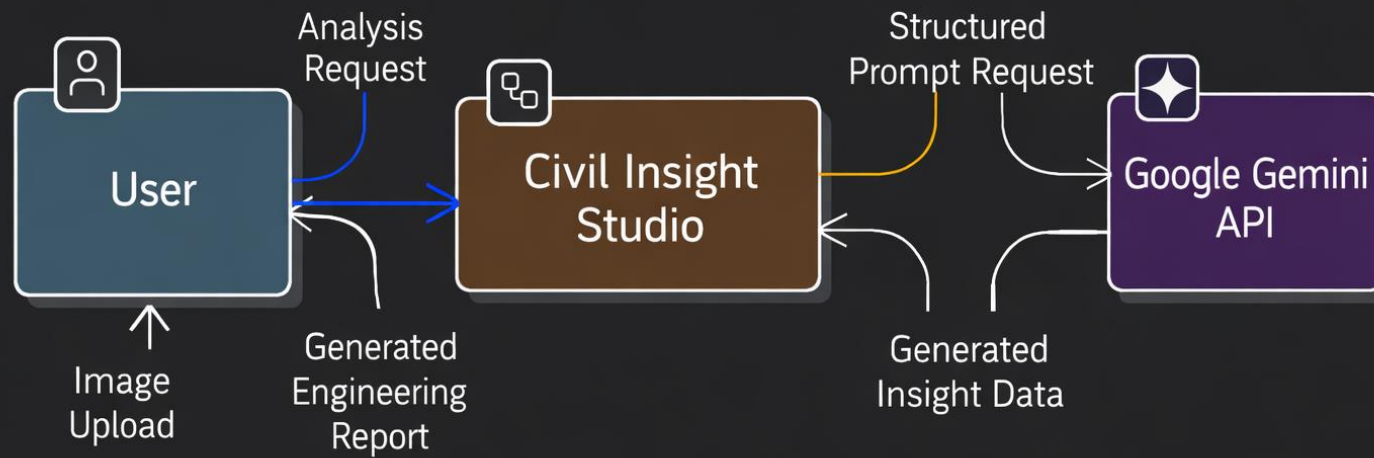
Date	3 February 2026
Team ID	LTVIP2026TMIDS24218
Project Name	Civil Engineering Insight Studio
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) visually represents how data moves through the Civil Engineering Insight Studio system. It shows how user input (text + image) is processed by the AI model and how structured engineering insights are generated and displayed.

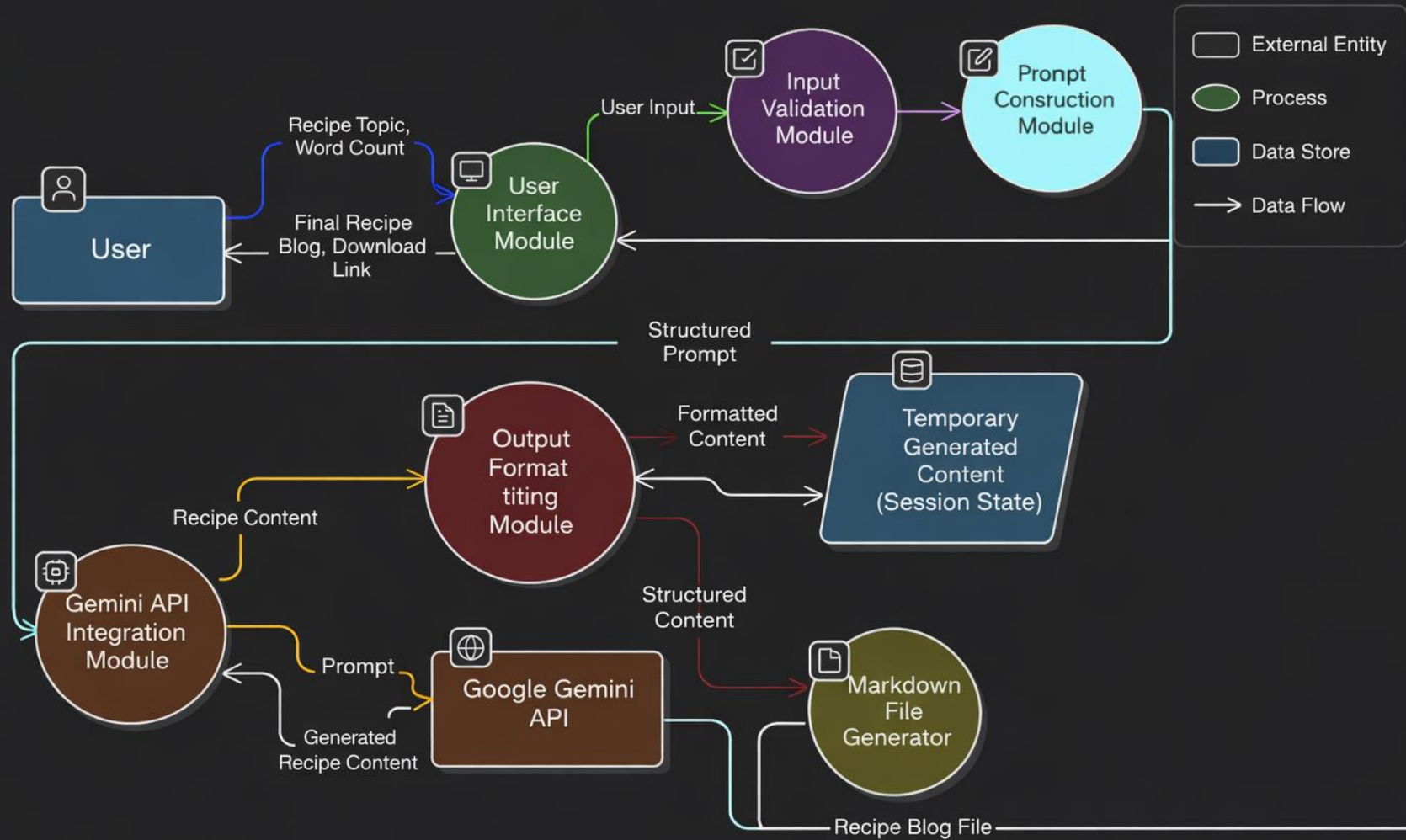
DFD Level 0 (Industry Standard)

Civil Engineering Insight Studio



DFD Level 1 (Industry Standard)

Civil Engineering Insight Studio



Civil Engineer (Web User)	Enter Analysis Request	USN-2	As a user, I can enter a structural analysis request.	System accepts text input and includes it in the AI prompt.	High	Sprint-1
Civil Engineer (Web User)	AI Structural Analysis	USN-3	As a user, I can generate a detailed structural report using AI.	System identifies materials, structural components, and construction stage.	High	Sprint-1
Civil Engineer (Web User)	Display Generated Report	USN-4	As a user, I can view the generated structural analysis directly on the screen.	Generated report appears clearly formatted in the UI.	High	Sprint-1
Civil Engineer (Web User)	Download Report	USN-5	As a user, I can download the generated structural report.	Clicking download provides a valid report file containing generated content.	High	Sprint-1

Civil Engineer (Web User)	Input Validation	USN-6	As a user, I am prevented from generating analysis without uploading an image.	System shows validation message and blocks analysis.	High	Sprint-1
Civil Engineer (Web User)	Regenerate Analysis	USN-7	As a user, I can regenerate analysis by modifying the prompt or uploading a new image.	New output replaces previous result without system error.	Medium	Sprint-2
System	Gemini API Integration	USN-8	As the system, I must send image + structured prompt to Google Gemini API and receive a response.	API request succeeds and returns structured engineering insights.	High	Sprint-1
System	Error Handling	USN-9	As a user, I receive an error message if AI analysis fails.	System displays clear error message without crashing.	Medium	Sprint-2